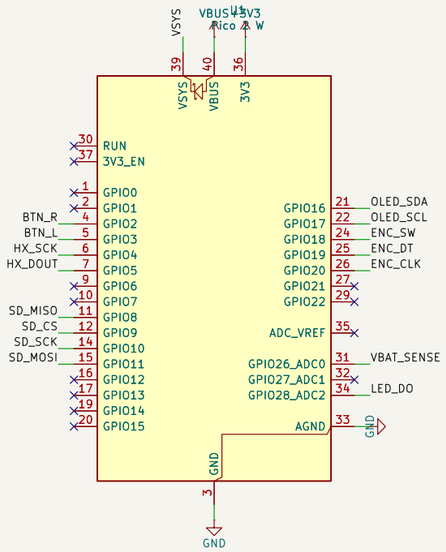
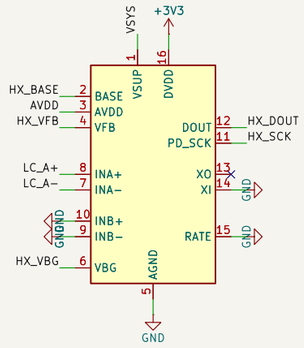
MixMeasureBuddy carrier board — Raspberry Pi Pico 2 W

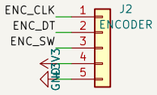
Pin mapping = src/firmware/src/config.py (OLED I2C0 GP16/17, buttons GP3/GP2, HX711 GP4/GP5, encoder GP20/19/18, WS2812 GP28, SD SPI1 GP8–11). Buttons use the internal pull-ups.



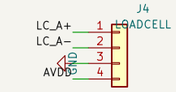
HX711: VSUP=VSYS, DVDD=3V3 (3.3 V logic), AVDD = 1.25 V x (R1+R2)/R2 = 3.75 V, E− tied to AGND, RATE=GND -> 10 SPS, internal oscillator (XI=GND).

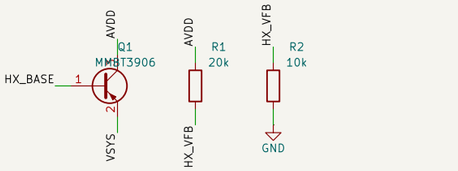


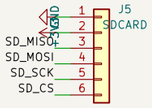


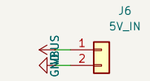




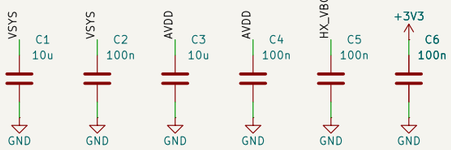


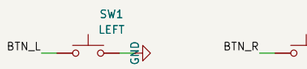








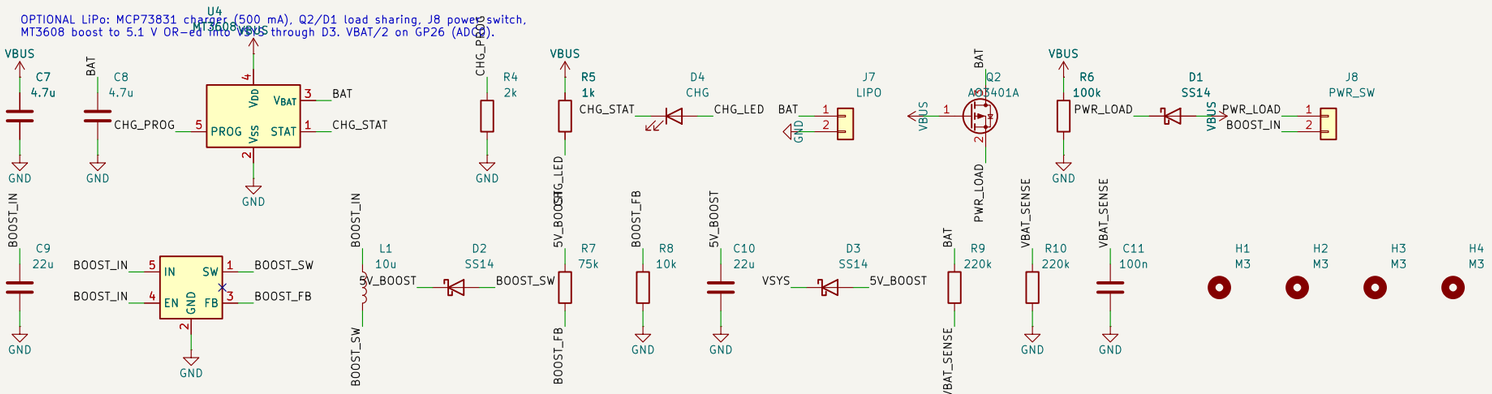




Load cell: J4 1=A+ (green) 2=A− (white) 3=E− (black) 4=E+ (red)



OPTIONAL LiPo: MCP73831 charges (500 mA), Q2/D1 load sharing, J8 power switch, MT3608 boost to 5.1 V OR-ed into VBUS through D3. VBAT/2 on GP26 (ADC0).



Sheet: /
File: mmb_carrier.kicad_sch

Title: MixMeasureBuddy Pico 2 W carrier

Size: A3	Date: 2026-10-31	**Rev: 1.0**
KiCad E.D.A. 10.0.8		Id: 1/1